

MATH 1342 – Elementary Statistics – Week 10: A Closer Look at Testing and Making Connections
Oct 27 – 29 | 4.4 A Closer Look at Testing • 4.5 Making Connections | Fall 2026 · Section 036 · TTh 6:00–7:20 PM (DIL)

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. Describe Type I and Type II errors in the context of a study, relate α to the probability of a Type I error, and explain the trade-off between the two errors and the effect of sample size. (CLO 4, CLO 8) 2. Explain how multiple testing and publication bias produce false positives, and distinguish statistical significance from practical significance. (CLO 8)	e-Text: 4.4 A Closer Look at Testing Interactive Lecture: 4.4 A Closer Look at Testing Slides: 4.4 A Closer Look at Testing (includes Khan Academy Type I/II video)	Participation: Errors in Testing – battery-life example (a)–(d): describe each error in context; courtroom analogy matching; vitamin E multiple-testing questions	Online Homework: 4.4 A Closer Look at Testing (a) (due Sun Nov 1, 11:59 PM)	WileyPLUS StatKey Blackboard Zoom (live class)
3. Describe how to generate randomization samples for a given study design (coin flips, re-randomizing groups, shifting data then bootstrapping). (CLO 8) 4. Compare bootstrap and randomization distributions by their center and use, identifying each center for given data. (CLO 5) 5. Use a 95% confidence interval to reach a two-tailed test conclusion at $\alpha = 0.05$, and decide whether a confidence interval or a hypothesis test fits a research question. (CLO 6, CLO 8)	e-Text: 4.5 Making Connections Interactive Lecture: 4.5 Making Connections Slides: 4.5 Making Connections	Participation: Making Connections – body temperature: test 98.6 using the CI (98.05, 98.47); sleep study (28/50): how to generate randomization samples and where each distribution is centered	Online Homework: 4.5 Making Connections (due Sun Nov 1, 11:59 PM)	WileyPLUS StatKey Blackboard Zoom (live class)
<i>Items that cover objectives from more than one row above (objectives in parentheses)</i>			Take-Home Quiz 6 (due Tue Nov 3, 11:59 PM) – covers 4.4, 4.5 (all objectives)	Assignment Tool in Blackboard
<i>Course resources and assessments of earlier weeks (not aligned to this week's objectives)</i>			Practice Test 2 (Blackboard, by Sun Nov 1) Test 2 – Fri Oct 30 – Sun Nov 1 (Blackboard, proctored, 2 hours). Covers Weeks 6–9.	Assessment Tool in Blackboard Respondus LockDown Browser and Monitor

CLO 4 Explain the role of probability in statistics. **CLO 5** Examine, analyze, and compare various sampling distributions for both discrete and continuous random variables. **CLO 6** Describe and compute confidence intervals. **CLO 8** Perform hypothesis testing using statistical methods.